



Helsinki Platform Display

Model No: DB052605 & DB053205



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TITLE

Operator and Maintenance Manual

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LIST OF ABBREVIATIONS

A	units - Amperes
AC	Alternating Current
Assy	Assembly
CIS	Customer Information System
COMMs	Communications
DC	Direct Current
F	Farhenheight
FET	Field Effect Transistor
GND	Ground
Hz	units - Hertz
in	units - Inches
Kg	units - Kilograms
lb	units - Pounds
LED	Light Emitting Diode
LK	Link
LSB	Least Significant Bit
mA	units - Milliamperes
mm	units - Millimetres
MON	Monitor Input
MSB	Most Significant Bit
N/A	Not Applicable
N/C	No Connection
PCB	Printed Circuit Board
PL	Plug
PSU	Power Supply Unit
QTY	Quantity per Display
Ref.	Refer to
RX	Receive
SBC	Single Board Computer
TB	Terminal Block
TX	Transmit
V	units - Volts
V AC	units - Volts (Alternating Current)
V DC	units - Volts (Direct Current)

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1 Introduction

1.1 Scope

The scope of this document covers the Type 1 and Type 2 LED passenger information displays for the Helsinki display project. The purpose of the document is to provide the relevant information to enable the user to install and maintain the displays correctly.

Because the two display types are similar, the Type 1 display is described throughout this manual.

1.2 Health and Safety

The following warnings and cautions are applicable to the display:

WARNING - HIGH VOLTAGE

THE DISPLAY CONTAINS MAINS VOLTAGES UP TO 240V AC. YOU MUST DISCONNECT ALL POWER FROM THE DISPLAY, THEN ALLOW AT LEAST ONE MINUTE TO ELAPSE BEFORE PERFORMING ANY MAINTENANCE. YOU MUST ALSO DISCONNECT POWER FROM THE MOUNTING POLE, THEN ALLOW AT LEAST ONE MINUTE TO ELAPSE BEFORE INSTALLING OR UNINSTALLING THE DISPLAY.

You can disconnect power by (in order of safety):

1. Unplug the external mains supply to the display at source.
2. Put the Circuit Breaker in the OFF position.

WARNING - HEIGHT HAZARD

THE DISPLAY IS DESIGNED TO BE MOUNTED AT THE TOP OF A CUSTOMER FURNISHED STRUCTURE, WHICH REPRESENTS A HEIGHT HAZARD WITH DANGER OF FALLING. YOU MUST CONSTRUCT A SCAFFOLDING TOWER OR USE A MEWP TO ENABLE THIS MOUNTING TO BE PERFORMED SAFELY.

CAUTION HEAVY EQUIPMENT

YOU MUST ENSURE THAT THE UNIT IS ADEQUATELY SUPPORTED DURING INSTALLATION, REMOVAL AND MAINTENANCE. ALL LIFTING AND CARRYING OF THE DISPLAY MUST BE PERFORMED BY AT LEAST TWO PEOPLE.

1.3 Description

Type 1 and *Type 2* single-sided metro displays are five line units with a clock at the top right corner. Each line on the *Type 1* display consists of a 160 wide x 8 high matrix while each line on the *Type 2* display consists of a 192 wide x 8 high matrix. The clock is a 48 x 8 pixel matrix on both display types. All displays use amber surface mount LED technology.

Displays consist primarily of three sections, the display housing, front door assy and the display plate assy.

The front door assy is locked with 4 tamper-proof, $\frac{1}{4}$ turn cam locks. A 7mm triangular key is required to open these locks. This door opens upwards and is held open using metal stays at either side. The front face on the door is a grey polycarbonate screen.

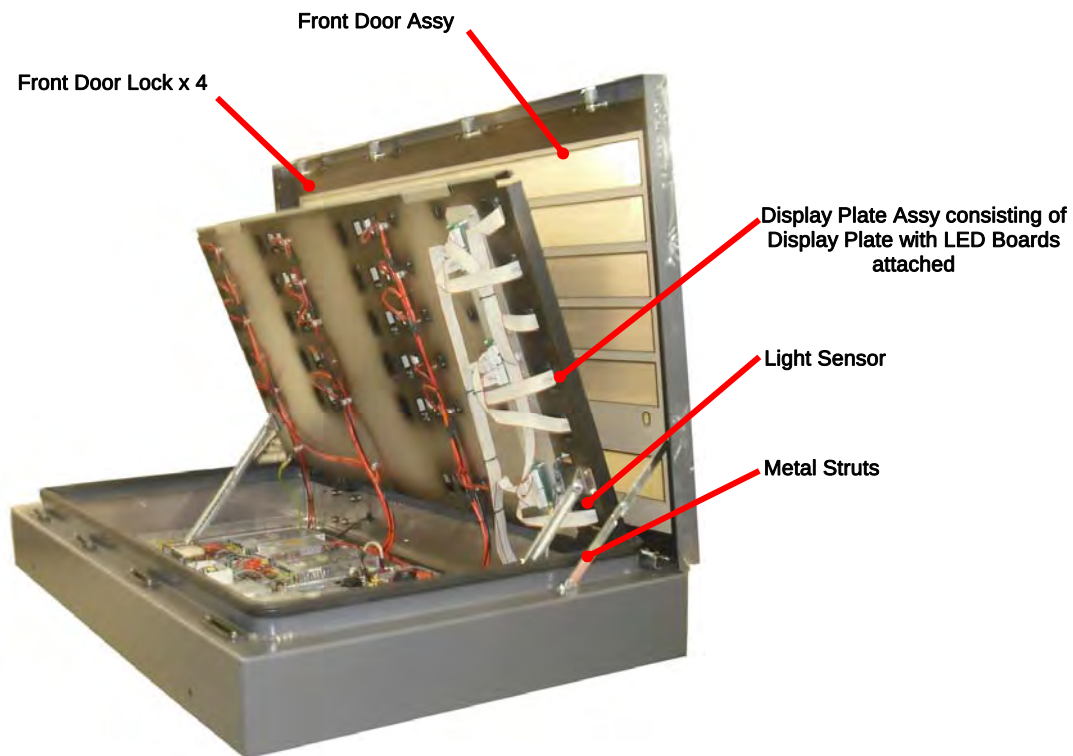


Figure 1.1 – View with Front Door Open.

The display housing is fabricated from 2mm aluminium and has a grey powder coated paint finish. The display plate assy is attached to the housing by hinges at the top and secured by magnetic fasteners at the bottom. This plate assy opens upwards and is held in the open position by metal stays at either side.

Attached to stand-offs on the display plate assy are the LED display boards.

Each display requires a mains power supply of 240V AC and COMMs source. The cables are routed through cable glands on the top of the display.

Displays also have the following additional features:

- Light sensor that controls the LED brightness depending on the ambient light levels.
- 10 Amp fuse circuit breaker to protect the display from current overload.
- PSU Monitoring.

No routine or periodic maintenance is required but information for fault finding and remedial maintenance is supplied.

1.4 Display Specifications

CHARACTERISTIC	VALUE
Display Features	
Models	DB052605, DB053205, Single Side
Display Type	Metro Type 1 & Type 2
Number of Lines	5 with Clock
Display Board Type	MB640805PLCC2 & MB480805PLCC2
LED Colour	Amber, Surface Mount
LED per Pixel	1
Pixel Pitch	7.62mm
Communication	
COMMs	Ethernet / RS232
Electrical	
Mains Power Supply	240V AC
Housing	
Dimensions (LxHxD)	Type 1: 1375mm x 800mm x 200mm Type 2: 1625mm x 800mm x 200mm
Enclosure	Single Sided
Access	Front, Removable Door Assembly
Housing Material	2mm Aluminium
Housing Colour	Grey, RAL 7024
Front Screen	Clear, Lexan Polycarbonate
Environmental Conditions	
Temperature Range	-13 degrees F to + 140 degrees F max (60 °C)

Table 1.1 – Display Specifications.

DISPLAY TYPE	POWER CONSUMPTION
DB052605 Type 1	320 Watts
DB053205 Type 2	280 Watts

Table 1.2 – Display Power Consumption.

2 Hardware Description

2.1 Display Boards, MB640805-PLCC2 & MB480805-PLCC2

The display face consists of MB640805-PLCC2 and MB480805-PLCC2 display boards.

The boards are driven in a multiplex mode with the row anodes connected to the horizontal drive and the column cathodes connected to the vertical drive. The boards contain the display driver circuitry, which generates the multiplexing signals for illuminating the LEDs.

Data enters each line through a 20-way screened ribbon cable from the logic board. The display boards require a 3.3V display power input, 5V logic power input and 20-way data/control input/output connections.

The display boards are fixed to stand-offs attached to the display plate door assy using M3 nuts.

NOTE 1: For data termination on each line, insert jumpers at links LK4 at the end of each line, left-hand side as viewed from front.

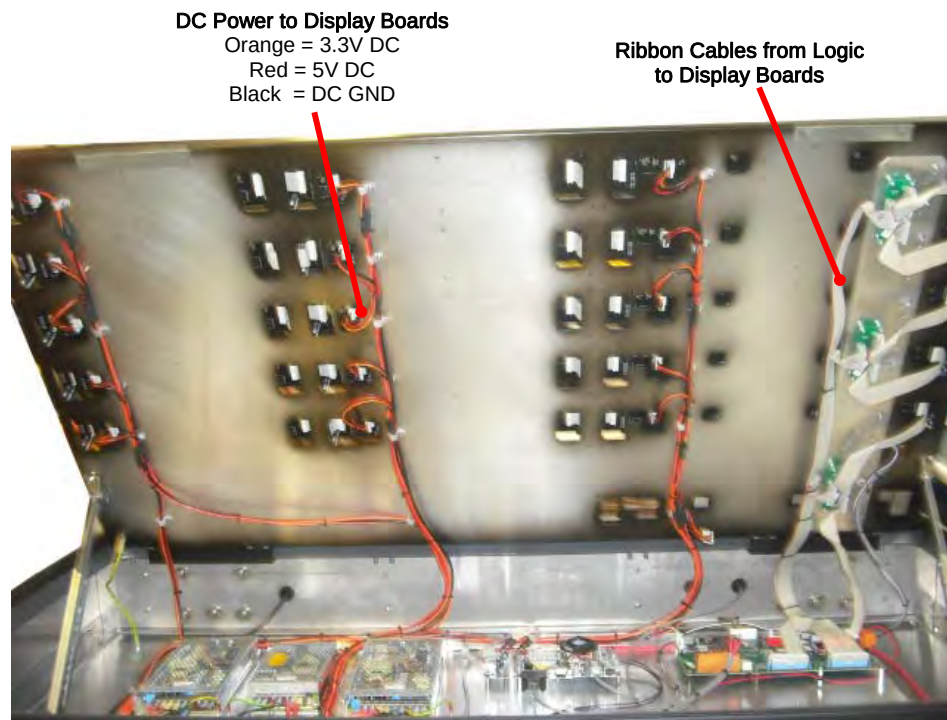


Figure 2.1 – Typical Display Board.

2.2 Logic Board, PCB236 / 237

The PAS236/237 board consists of a V53 Micro Processor running at 29 MHz. (bus speed 14.5 MHz.) and associated peripheral devices together with the following features necessary to control and monitor the Display System as a whole:

- o 4-way power input connector for 5V, 12V and two 0V inputs.
- o Two 40-way connectors and buffer/filter circuitry for the display interface signals consisting of SCLK, RD0-3, GD0-3, A, B, C, D, LE/MOD, OE/ED, RD_RTN0-3, GD_RTN0-3. Other signals on the headers are 15V FET drive supply and 0V signal returns
- o 512K x 16bit CMOS 5.0V low power SRAM memory, K6X8016C3B-F (Industrial grade)
- o 512K x 8bit/256K x 16bit CMOS 5.0V boot (top boot) sector flash memory, Am29F400B
- o On-board switches for Display configuration settings where necessary

Incoming RS232 signal wires from the network are terminated at the RS232 channel at PL20.

5 Volt Logic power and the 12 Volt power used for the gate control of the row driver FET's on the Display Boards, enter the Logic Board by way of a screw terminal block at PL5.

At PL12 are the Power Supply Failure signals from each 3.3V PSU in the Display.

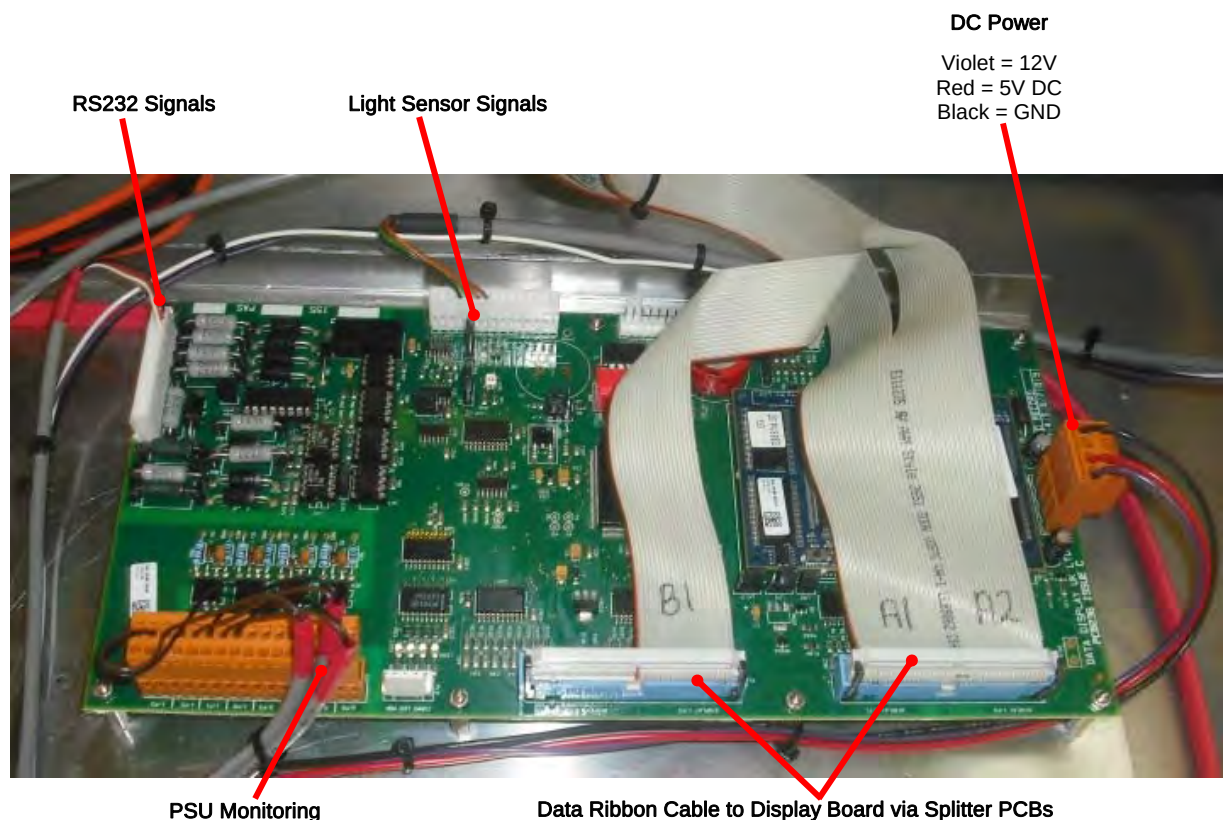


Figure 2.2 – PCB236/237 Logic Board.

2.2.1 Dip Switch Settings

The dip switches, SW1 and SW2, on the logic board consists of a bank of 8 switches that allows the user to:

- Set the display address
- Set Local Loader Mode

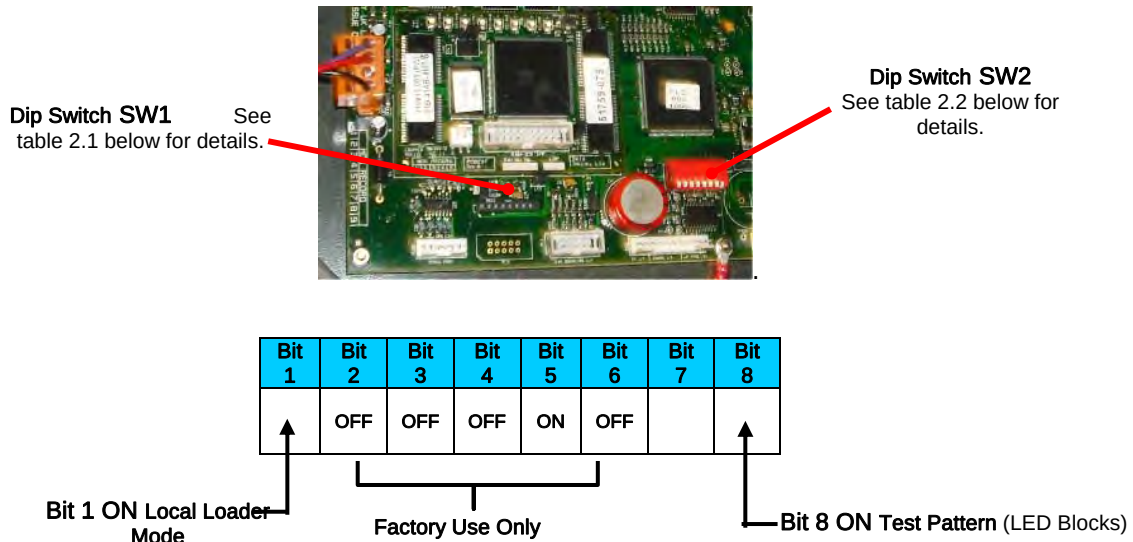


Table 2.1 – Dip Switch SW1 Functions.

Sw 1 Bit	Sw 2 Bit 7	Sw 3 Bit 6	Sw 4 Bit 5	Sw 5 Bit 4	Sw 6 Bit 3	Sw 7 Bit 2	Sw 8 Bit 1	Function Addressing Base Address = 40 hex
Baud Rate	OFF	OFF	OFF	OFF	OFF	OFF	ON	40h + 01h = Address 41 hex (65 dec) (Char 'A')
Baud Rate	OFF	OFF	OFF	OFF	OFF	ON	OFF	40h + 02h = Address 42 hex (66 dec) (Char 'B')
Baud Rate	OFF	OFF	OFF	OFF	OFF	ON	ON	40h + 03h = Address 43 hex (67 dec) (Char 'C')

SW 2 – Sw 1 = Baudrate Control

*ON = 19200 baud
OFF = 9600 baud*

Table 2.2 – Dip Switch SW2 Functions.

DEFAULT OPERATION:

SW1 = All OFF

SW2 = Sw1 = On, Sw8 = On all others = Off => 81 hex (19200 baud and Address '0A')

2.3 Single Board Computer (ECM-A50M)

The ECM-A50M SBC is a 3.5" FDD size single board computer (SBC) that comes with a processor and various interfaces that allow the information to be extracted from the customer's data feed. The customer's data feed is connected to the RJ45 Network Socket LAN 1 port on the SBC.

The Single Board Computer has the following features:

- AMD eOntario T40E 1.00GHz CPU
- AMD A50M Chipset
- One 204-pin DDR3 SODIMM Socket Supports up to 4GB DDR3 1066 SDRAM
- Dual View, 2-CH LVDS, CRT, HDMI
- 701-CH Audio, Dual Gigabit Ethernet
- 1 CF, 2 SATA, 2 COM, 7 USB, 16-bit GPIO
- Power Requirement +12V
- Operating Temperature: 0°C ~ 60°C
- Power Consumption 2.1 A 25.2 Watts

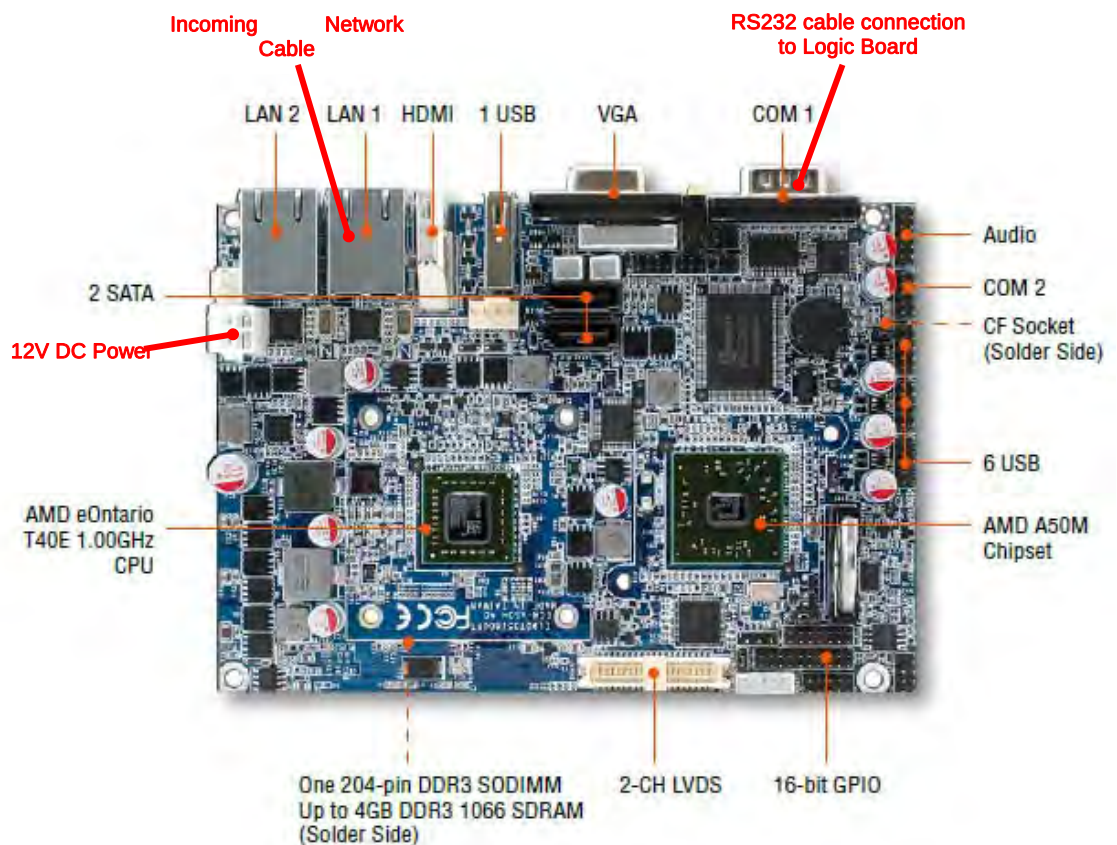


Figure 2.3 – ECM-A50M SBC.

2.4 LDR Board (Light Sensor), PAS175DD

The light sensor attached to the display plate assembly controls the brightness of the LEDs on the display. This circuit utilizes a photodiode detector. The output is converted using an analog to digital converter. The logic board monitors the digital output and the luminosity of the LEDs on the display is adjusted accordingly.

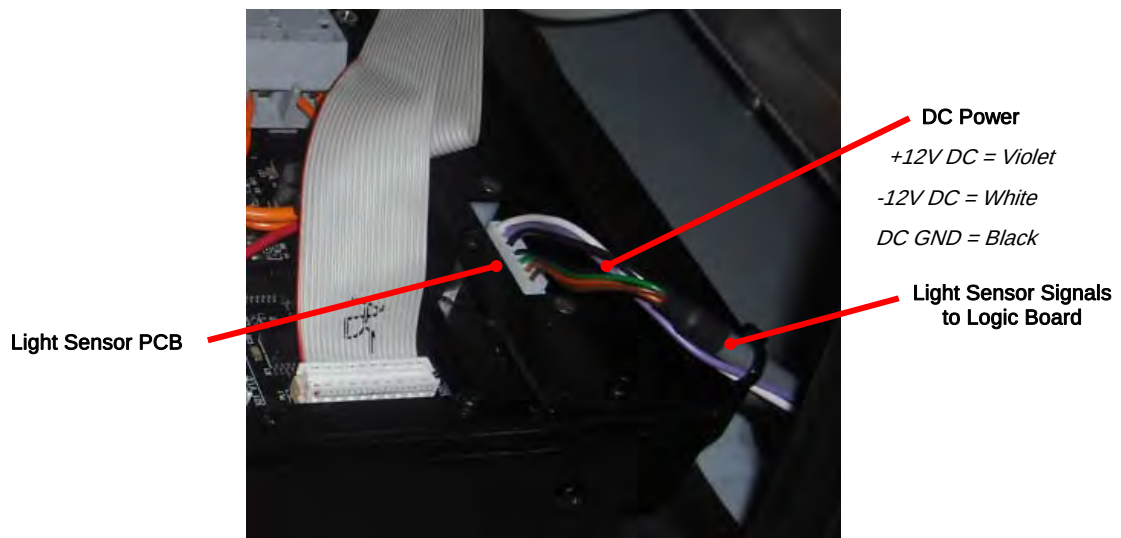


Figure 2.4 – Light Sensor, PAS175DD.

For an operational check, place your hand over the sensor and observe the difference in the brightness of the LEDs.

2.5 Splitter Board, RG SPLITTER PCB

The RG SPLITTER V1.0 splits the incoming 40-way ribbon cable into two outputs. The incoming signal terminates at PL1. The 20-way cables labelled A1, A2 and B1 bus data to display boards via the RG splitter PCBs.

2.6 Relay Switch

A customer furnished relay switch is installed in the display. The relay switch allows the user to power up or down the display remotely via IP through a web browser.

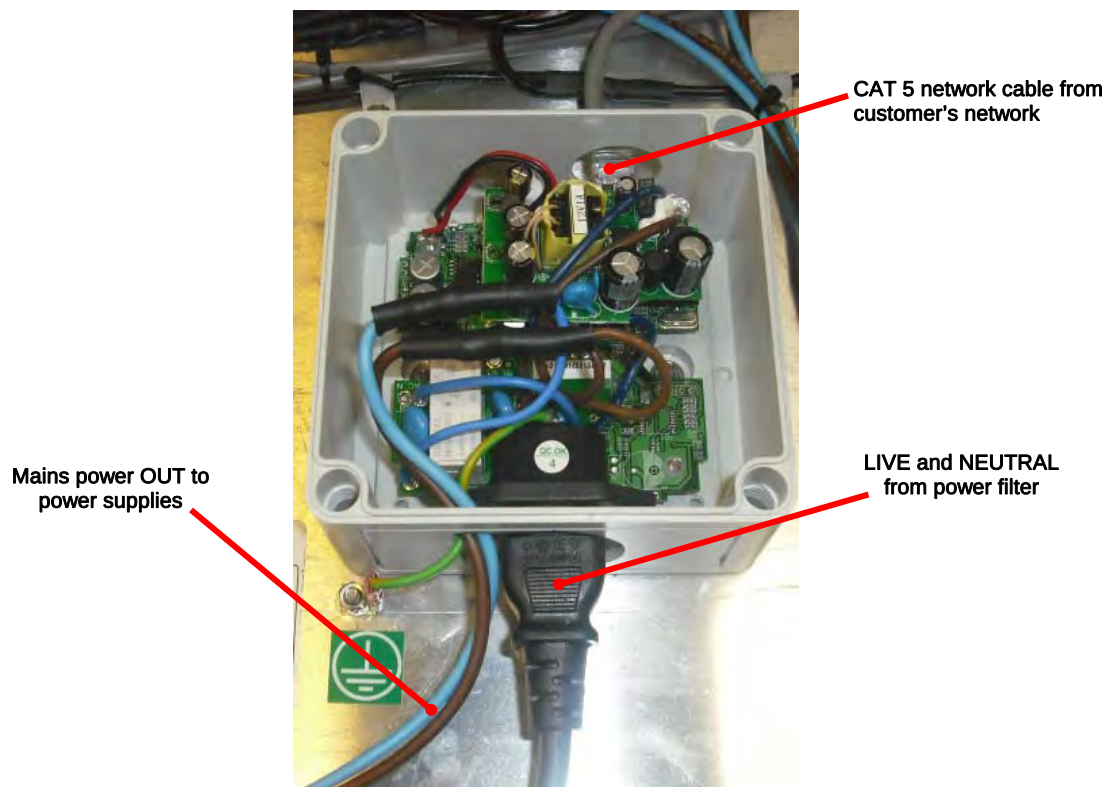


Figure 2.5 – Relay Switch.

3 Electrical

3.1 Description

The displays operate from a mains power supply of 240V AC. A 10 feet long external mains power fly cable is supplied with the display.

The power supply units (PSU) mounted inside on the display chassis convert the incoming AC power to DC power.

The LIVE and NEUTRAL conductors are filtered and connected to the relay switch via a 10 amp dual-pole type circuit breaker. The LIVE and NEUTRAL source for the PSUs is then taken from the relay switch. The LIVE, NEUTRAL and EARTH wires are routed as shown in the figure below. For further wiring, refer to wiring diagrams in chapter 7.

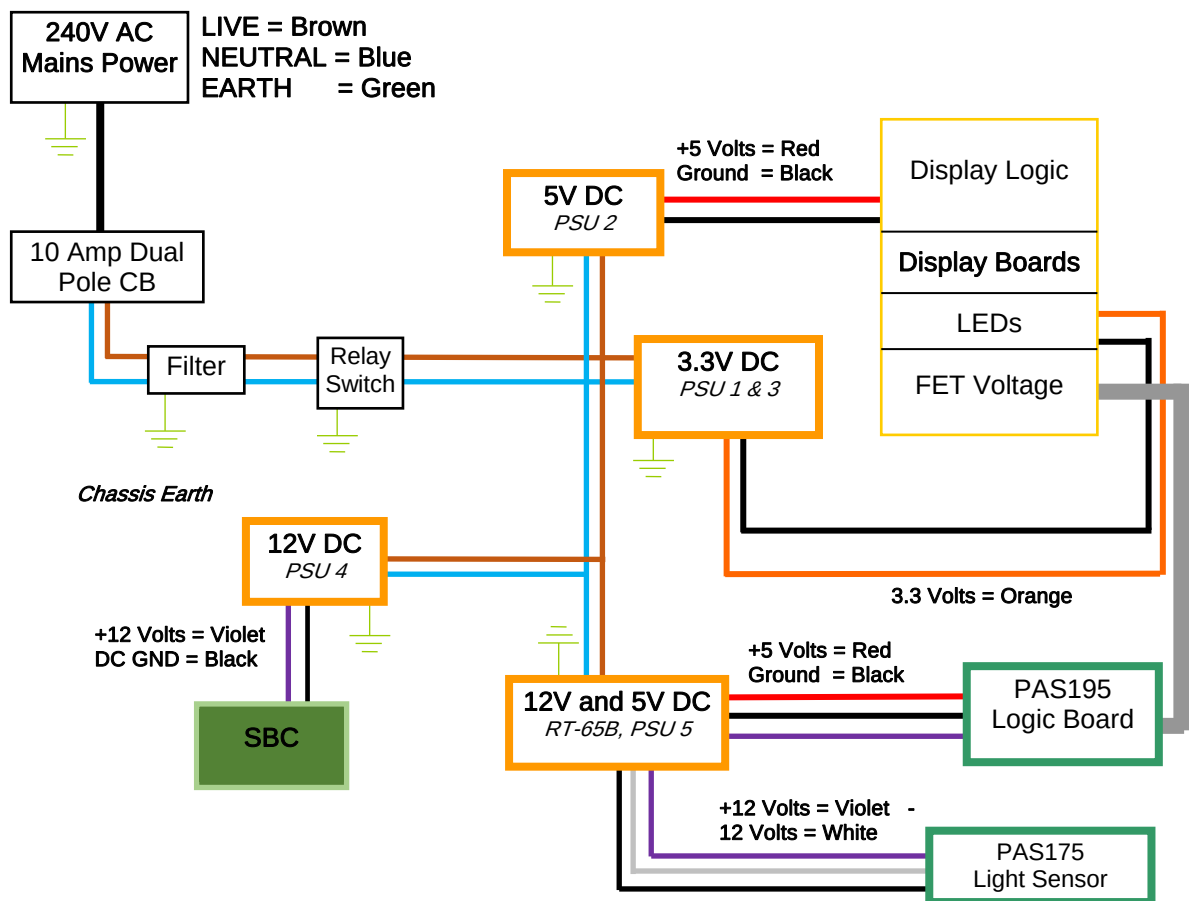


Figure 3.1 – Power Supply Block Diagram.

3.2 DC Power Supply

There are four types of power supplies units (PSU) that convert the AC mains power to DC power installed in the display:

- 3.3V 200 Watt PSU, 2 off, labelled PSU 1 & 3
- 5V 100 Watt PSU, 1 off, labelled PSU 2
- 12V 50 Watt PSU, 1 off, labelled PSU 4
- RT-65B PSU, 1 off, labelled PSU 5

3.2.1 Power Supply Unit (PSU No. 1 & 3), RSP-200-3.3

The RSP-200-3.3 PSU provides the 3.3V DC power to the all LED display boards and has the following features:

- International AC input range (115/230V) with selector switch
- Built in filter
- low ripple noise
- high efficiency
- short circuit protection
- overload protection
- over voltage protection



Figure 3.2 – Typical 3.3V Volt PSU Wiring.

The mains cables have crimp receptacles and are connected to the relevant LIVE, NEUTRAL and EARTH inputs on the screw terminal barrier strip.

3.2.2 Power Supply Unit (PSU No. 2), RS-100-5

The RS-100-5 PSU provides the 5V DC power to the all LED display boards and has the following features:

- International AC input range (115/230V) with selector switch
- Built in filter
- low ripple noise
- high efficiency
- short circuit protection
- overload protection
- over voltage protection



Figure 3.3 – Typical 5V Volt PSU Wiring.

The mains cables have crimp receptacles and are connected to the relevant LIVE, NEUTRAL and EARTH inputs on the screw terminal barrier strip.

3.2.3 Power Supply Unit (PSU No. 4), S-50-12

The S-50-12 PSU provides the 12V DC power to the single board computer (SBC) and has the following features:

- o 12 Volt output at 3.5A
- o International AC input range
- o Built in filter
- o low ripple noise
- o high efficiency
- o short circuit protection
- o overload protection
- o over voltage protection



Figure 3.4 – Typical 12V PSU Wiring.

The mains cables have crimp receptacles and are connected to the relevant LIVE, NEUTRAL and EARTH inputs on the screw terminal barrier strip.

3.2.4 Power Supply Unit (PSU No. 5), RT-65B

The RT-65B PSU provides the 12V DC power to the logic board, light sensor PCB and Serial Server. And 5V DC power to the display boards. This PSU is a multi-output 65W AC/DC power supply and has the following features:

- o 5 Volt output at 8A
- o 12 Volt output at 3.5A
- o -12 Volt output at 0.5A
- o International AC input range
- o Built in filter
- o low ripple noise
- o high efficiency
- o short circuit protection
- o overload protection
- o over voltage protection



Figure 3.5 – Typical RT-65B PSU Wiring.

The mains cables have crimp receptacles and are connected to the relevant LIVE, NEUTRAL and EARTH inputs on the screw terminal barrier strip.

3.3 DIN Rail Assy

The DIN Rail Assy distributes the incoming AC supply to the filter and PSUs via a 10A circuit breaker.

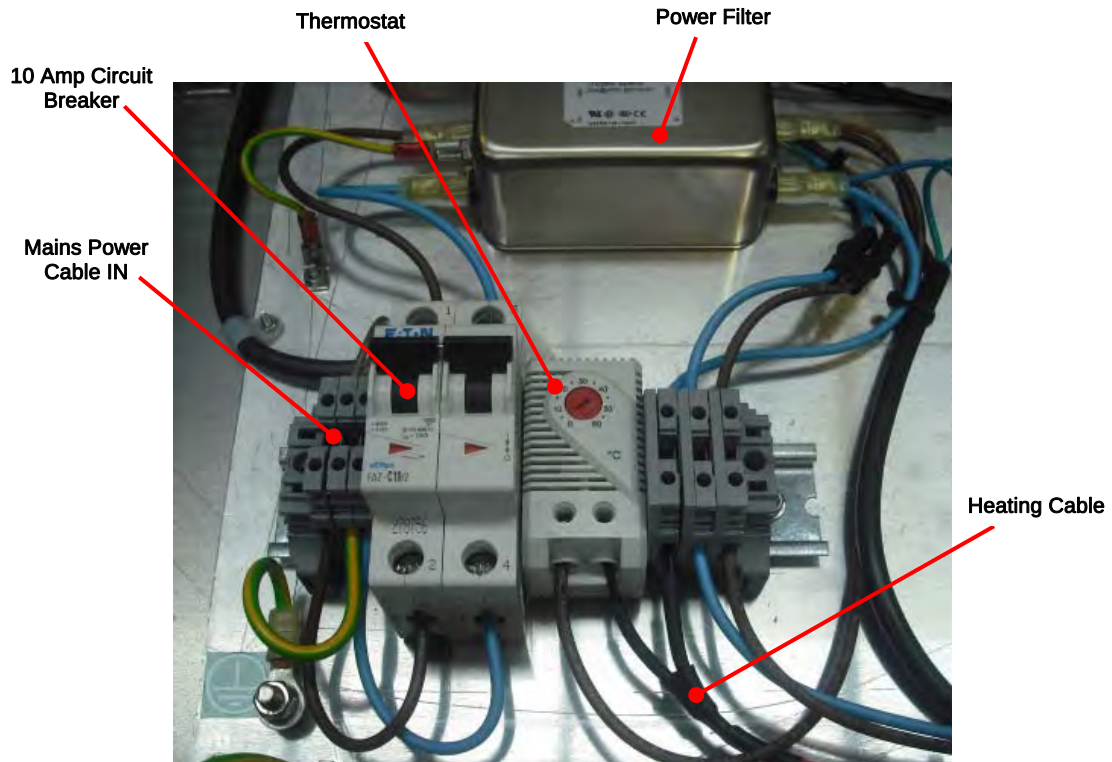


Figure 3.6 – DIN Rail Assy Layout.

3.4 Filter

The mains power filter is a 20 Amp multistage general purpose filter with two inductors per phase for high common mode attenuation.

3.5 Circuit Breaker

The dual pole, 10 Amp, circuit breaker is located on the DIN Rail assy. This circuit breaker fuses the incoming LIVE and NEUTRAL conductors that power the display.

4 Installation

This chapter details the tasks required to complete the installation of a display.

4.1 Precautions

WARNINGS

Before undertaking the above tasks, note the following:

1. Displays are powered from a mains power supply of 240V AC. Before carrying out any installation work involving 240V AC mains, make sure that all connection points are dead, i.e. isolated from the source of the mains power supply. Verify the points to be worked on are dead by using appropriate test equipment.
2. All installation personnel must be familiar with good manual lifting practice. Appropriate safety measures shall be enforced while work is carried out and in particular special precautions taken when working from ladders and/or scaffold platforms. Lifting equipment must be used to position the display just prior to fixing into position at the site or if the displays are to be moved over long distances.

REGULATIONS

All personnel involved in carrying out the work must be aware of any site working regulations for the contract and must have certification for any formal training required.

Before you do any installation or maintenance task, make sure that the following safe working practices are adhered to:

- Only allow qualified personnel to do maintenance tasks
- Provide sufficient illumination for the task, especially during non-daylight hours
- Wear personal protective equipment relevant to the maintenance task
- Use insulated tools and equipment when working near or with energized parts
- Take measures to avoid inadvertent contact of conductive materials or equipment with energized parts of the display
- Take measures to establish safe working distances from energized parts



ESD PROTECTION

- When working within the display electrostatic device, protection measures must be used as there are a number of electronic assemblies which are static sensitive and must not be touched unless suitable grounding wristbands are used.
- A wristband and lead can be obtained from RS Components, Stock No. 287-7385, which will clip to the EBP studs.

4.2 Mounting the Display

Tools and Equipment

The following tools and equipment are required:

- Suitable raised platform/scaffolding or Mobile Elevating Work Platforms (MEWP)
- Mounting hardware, brackets, fasteners etc

Procedure

There are M8 threaded nutserts at locations on the top of the display that allow the mounting brackets to be attached to the display.

Care should be taken so as not to damage the electronic assemblies inside when moving or mounting the display.

Please see figure below for nutsert locations.

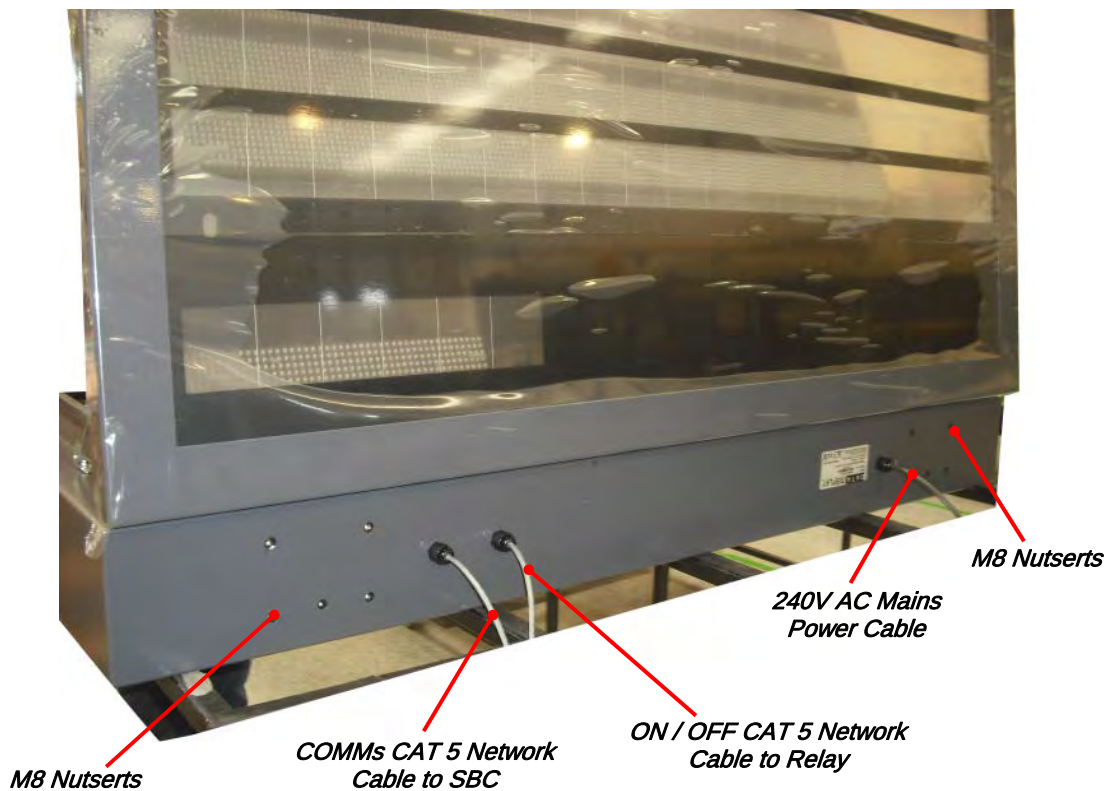


Figure 4.1 – Mounting Points.

4.3 Opening the Display

The display front door is hinged on its top edge and is held closed using security locks along the bottom edge and side of the display:

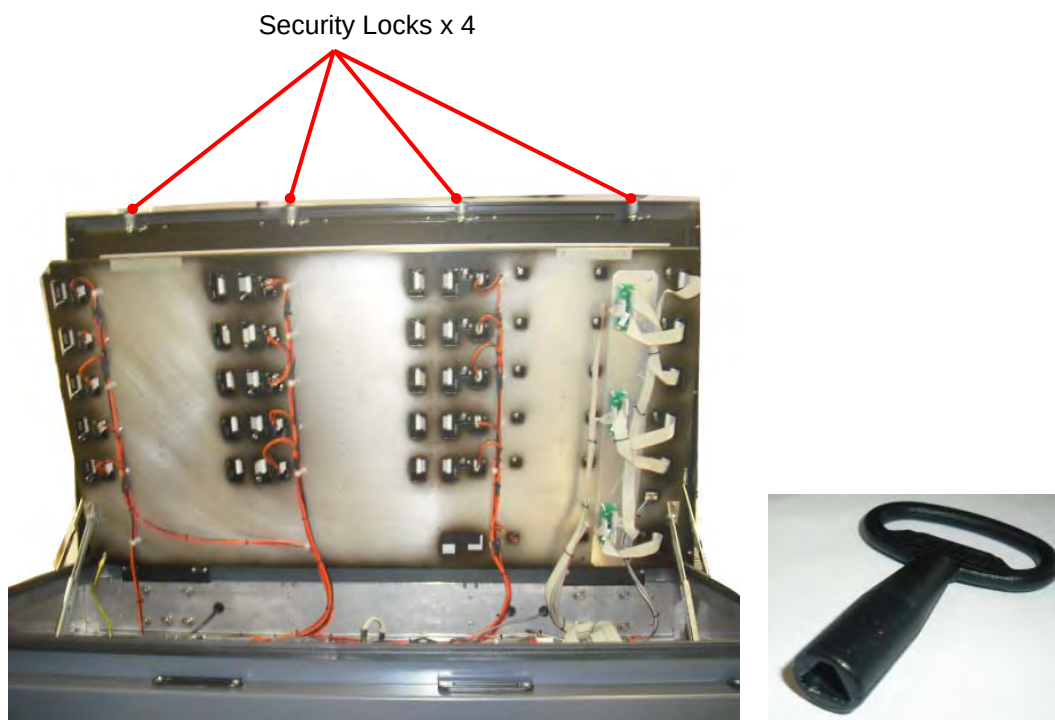


Figure 4.2 – Location of Locks and Key.

To open the display, do the following steps:

- a) Insert the key in the locks and turn clockwise to the open.
- b) Carefully lift the front door assy upwards until the metal stays are in the locked position. **Make sure that the metal stays are fully extended and in the locked position before releasing the door assy.**
- c) Carefully lift the display plate assy upwards fully and lower slowly until the ratchet stays are in the locked position.
- d) Do the steps in reverse order to close the display.

4.4 Mains Power Requirements

An external 240V AC supply is required to power the display. The customer supplies the mains or equivalent supply cable.

It is **UNSAFE** to do maintenance tasks when the mains power is **ON** within the Display.

Pull the 2-pole switch downwards to isolate the mains power within the Display.

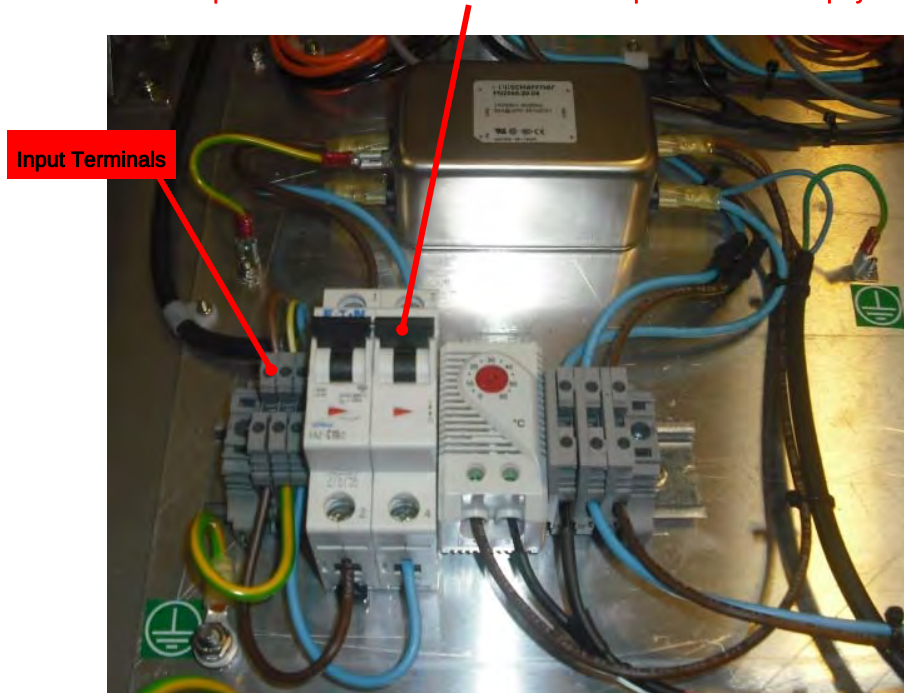


Figure 4.3 – DIN Rail Assy.

Important cable stripping note, if required - Only strip back enough of the mains cable sheath and braid to allow the break out of the wires for connection to the input terminals on the DIN rail assy.

The external mains cable wires terminating in the display should be connected as follows:

LIVE wire (Brown) to LIVE terminal

NEUTRAL wire (Blue) to NEUTRAL terminal

EARTH wire (Green) to EARTH terminal

A suitable tester should be used to verify that the mains cabling is connected correctly between local 240V AC mains power outlet and the display.

4.5 Communications

A vacant node on the customers Local Area Network (LAN) is required to provide communications to the display.

The on-board computer (SBC) is the communications interface that connects the incoming cable from the customers' network to the image controller in the display. The incoming CAT 5 cable is routed through the cable entry gland at the left side of the display.



Connect Incoming CAT 5 Ethernet
Cable Network here

Figure 4.4 – Network CAT 5 Cable Connections.

5 Fault Finding and Diagnostics

5.1 Description

When a fault is reported, check the management software to identify if the log for the faulty display has highlighted the fault.

Faults include:

- Total display failure (5.1.1)
- Partial display failure (5.1.2)
- Single display board failure (5.1.3)

Refer to **Chapter 6** for removal and installation procedures for replacement parts.

Reported Fault	Possible Faults	Repair
5.1.1 Total display failure.	A) COMMs connections	A) Check that communications with the display is OK on the management software. Read the logs for the faulty display and check if any there are any failures listed.
	B) External power is disconnected or Circuit Breaker is switched OFF.	B) Connect the external power cable to the display is connected. Check that the Circuit Breaker is switched ON.
	C) One of the following components may be defective: SBC, Logic Board or Mains Filter.	C) These components have no defined lifetime span. If, however a component is defective, replace the component item as described in section 6.4 .

Reported Fault	Possible Faults	Repair
5.1.2 Partial display failure. E.g. A line of display boards are blank or showing spurious information.	A) Loose ribbon data cables between boards.	A) Check that the ribbon data cables are connected securely. Remove and replace faulty ribbon cable.
	B) No DC power at the DC connector.	B) Check that the DC power cables are connected securely. Do a visual check and check that the green LED on the PSUs is ON. If the green LED is not ON or flickering then the PSU is faulty. Replace the faulty PSU as described in section 6.4.1 . If the green LED is ON, replace the display board as described in section 6.4.3 .
	C) Faulty data cable from logic board.	C) Change the data cable from the logic board to the display board with another cable to confirm ribbon cable is OK. If fault is still, contact the Data Display Ltd technical support department, section 5.2

Reported Fault	Possible Faults	Repair
5.1.3 Single display board failure.	A) Single or cluster of LEDs blown.	A) If a single LED or small cluster of LEDs are faulty, replace the display board as described in section 6.5.5 .
	B) Loose ribbon data cables.	B) Check that the ribbon data cables are connected securely. If the board is still faulty, remove and replace faulty ribbon cable.
	C) No DC power at the DC connector.	C) Check that the DC power cables are connected securely. Do a visual check and check that the green LED on the PSUs is ON. If the green LED is not ON or flickering then the PSU is faulty. Replace the faulty PSU as described in section 6.4.1. If the green LED is ON, replace the display board as described in section 6.4.3.

For Technical Support Dept. Refer to Section 5.2.

Table 5.1 – Fault Finding.

5.2 Technical Support

For technical support and spare parts, contact your Daktronics customer service department.

Before contacting Daktronics, identify these important part numbers:

Display Serial Number:	
Display Model Number:	
Contract Number:	
Date Installed:	
Location of Display:	
Daktronics Customer ID Number:	

To call Daktronics customer service:

Tel #: (800)-833-3157

E-mail: TransportationCSR@Daktornics.com

6 Maintenance Tasks

6.1 Precautions

WARNINGS

Before undertaking the above tasks, note the following:

1. Displays are powered from a mains power supply of 240V AC. Before carrying out any installation work involving 240V AC mains, make sure that all connection points are dead, i.e. isolated from the source of the mains power supply. Verify the points to be worked on are dead by using appropriate test equipment.
2. All installation personnel must be familiar with good manual lifting practice. Appropriate safety measures shall be enforced while work is carried out and in particular special precautions taken when working from ladders and/or scaffold platforms. Lifting equipment must be used to position the display just prior to fixing into position at the site or if the displays are to be moved over long distances.

REGULATIONS

All personnel involved in carrying out the work must be aware of any site working regulations for the contract and must have certification for any formal training required.

Before you do any installation or maintenance task, make sure that the following safe working practices are adhered to:

- Only allow qualified personnel to do maintenance tasks
- Provide sufficient illumination for the task, especially during non-daylight hours
- Wear personal protective equipment relevant to the maintenance task
- Use insulated tools and equipment when working near or with energized parts
- Take measures to avoid inadvertent contact of conductive materials or equipment with energized parts of the display
- Take measures to establish safe working distances from energized parts

6.2 First Line Maintenance

6.2.1 Preventative Maintenance

Inspect the display weekly for defects. Do a visual check for the following:

Fault	Action
Check that there are no marks, scratches or cracks on the screen on the door assembly.	Use a suitable cleaning solution to remove the marks.
Observe that the display is receiving real time messages from the station management system.	If not, report fault to station manager.
Check that the door assembly is secured tightly.	If not, insert key and turn clockwise to lock.
Check that the cable glands are secured tightly.	If loose, twist the gland nut to lock.

Table 6.1 – Preventative Maintenance.

6.2.2 Metal Coating on Stays

During each maintenance visit, the coating on the mechanical stays should be observed. Each strut should be coated at each visit using a suitable greasing agent.

6.2.3 Tools

The following tools are required:

- Basic toolkit
- ESD Strap
- Calibrated Digital Multimeter (DMM) with Temperature Probe, e.g. TENMA 7275 available from Farnell
- Solder iron
- Spare ribbon cables
- Triangular locking key for doors

6.2.4 Cleaning the Display

Tools and Equipment

The following tools and equipment are required:

- Suitable raised platform/scaffolding or Mobile Elevating Work Platforms (MEWP)
- Bucket with warm water with soapy detergent
- Sponge
- Towel

Procedure

Routine cleaning of the exterior of the display is at the discretion of the maintainer and subject to local conditions.

Use non-abrasive materials such as a sponge or lint-free cloth, together with water mixed with a soft detergent (e.g. liquid soap) to clean the front screen.

Fill your bucket with warm water and add your cleaning solution.

Using the sponge or lint-free cloth, wash the front screen with the water-and-cleaner mixture, carefully removing any visible dirt and grime. Gentle pressure may be necessary.

Use a suitable cleaning solution to remove more difficult marks.

DO NOT USE sharp or metal objects.

Dry the screen gently with a towel. Make sure that there are no visible soapy streaks remaining on the front screen.

Caution: Do not use sharp or abrasive objects or cleaning materials, as these will damage the display

6.3 Line Replaceable Units

The following is a list of line replaceable units (LRUs) contained in a Display. Please cross reference with figure A1 in chapter 7 of this manual.

Type 2 display board quantities in brackets.

ITEM NO.	DESCRIPTION	PART NUMBER	QUANTITY
1	Circuit Breaker, Dual Pole, 10 Amp	S-3230168	1
2	Filter, Schaffner FN 2060-20-06, 20A, 2 stage	Z-3226895	1
3	Power Supply Unit, RS-200-3.3, 3.3V	A-3232097	2
4	Power Supply Unit, RS-100-5, 5V	A-3227819	1
5	Power Supply Unit, RS-50-12	A-3227942	1
6	Power Supply Unit, RT-65B, 5V, +12V, -12V	A-3227918	1
7	LED Display Board, Amber, MB640805PLCC2	0A-1936-0343	5 (15)
8	LED Display Board, Amber, MB480805PLCC2	0A-1936-0344	11 (1)
9	Logic Board, ARM 9, PCB236/237	0A-1940-0015 / 0047	1
10	Light Sensor Board	0A-1939-0007	1
11	SBC - 3.5" A50M with T56 processor, ECM- A50M-56N	A-3230313	1
12	Splitter PCB	0A-1939-0022	3
13	Thermostat KTO-1140 Red	S-3230042	1
14	Heating Cable, 230 Volt	W-3228291	1
-15	Key, 7mm	HS-3228914	1
-16	Brackets, Type 3	0L-3326333	2

Table 6.2 – Line Replaceable Units.

6.4 Removal and Installation Instructions

To access the electronic components, open the Door Assy using the Key supplied with the Display.

When removing and installing components, do the following steps:

1. Disconnect the external mains power supply to the display.

WARNING: After isolating power from the display, wait at least one minute before performing the next step.



WARNING: Some sub-assemblies are part of the mains AC power circuitry of the display. Danger of Electrocution.

2. Unlock the front Door Assy. Please refer to section 4.3. **Hold the front Door Assy firmly and lift upwards until the metal struts are fully extended. Lower the Door Assy slowly until the metal strut engages in the lock position.** Hold the inner Display Plate Assy at the bottom and lift upwards. Lift upwards until the metal struts are fully extended.
3. For safety, power-down the display. To power-down the display, put the circuit breaker switch in the OFF position.

It is UNSAFE to do maintenance tasks when the mains power is ON within the Display.

Pull the 2-pole switch downwards to isolate the mains power within the Display.

Power is ON when switch is in UP position

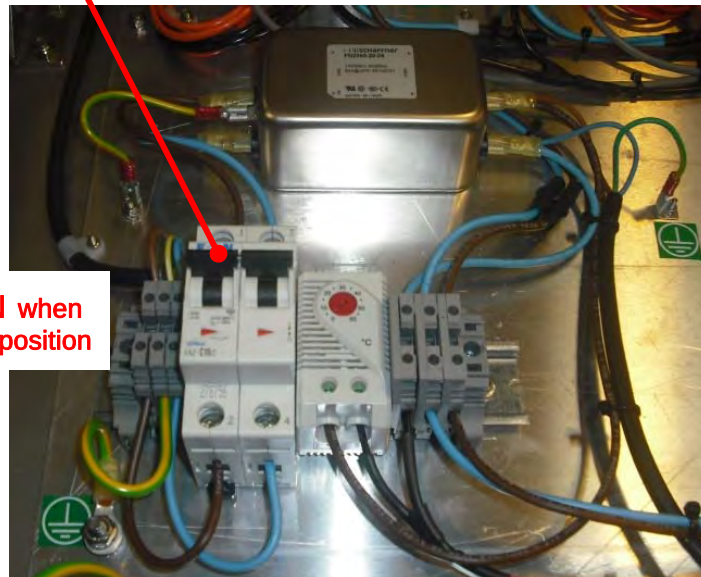


Figure 6.1 – Circuit Breaker Switch.

WARNING: After isolating power from the display, wait at least one minute before performing the next step.

4. **NOTE:** Retain all hardware removed from the display and store in a safe place.

**ESD PROTECTION**

When working within the display electrostatic device, protection measures must be used as there are a number of electronic assemblies which are static sensitive and must not be touched unless suitable grounding wristbands are used.

A wristband and lead can be obtained from RS Components, Stock No. 287-7385, which will clip to the EBP studs.

6.4.1 Removing a PSU

- a) Refer to **6.4.** and do steps 1 to 4.
- b) Make sure that there is no mains power at the LIVE terminal on the filter using a DMM.
- c) Locate the PSU to be removed and remove the clear plastic cover from the terminals, if necessary.
- d) Disconnect the crimp terminals and make note of their locations.
- e) Unscrew the fasteners that attach the PSU and remove the PSU. Retain all screws, washers, nuts etc. for further use.

6.4.2 Installing a PSU

- a) Spread a layer of heat sink compound over the rear of the new PSU. Attach the PSU to the display chassis using the fasteners retained during removal.
- b) Connect the crimp terminals.
- c) Replace the clear cover over the terminals, if necessary.
- d) Connect the mains power to the display at the external isolator.
- e) Check the DC output of the PSU.
- f) Test the display from the CIS.
- g) Clean the display after all tests are complete and fill in the maintenance report, if required.

6.4.3 Removing a Display Board

- a) Refer to **6.4.** and do steps 1 to 4.
- b) Make sure that there is no mains power at the LIVE terminal on the filter using a DMM.
- c) Locate the faulty display board.
- d) Unscrew the screws that attach the display board. Keep in a safe place.
- e) Disconnect the data and power connectors from the defective board and note their locations.
- f) Remove the defective display board.

6.4.4 Installing a Display Board

- a) Connect the data and power connectors in their correct locations on the new display board.
- b) Attach the board to the door assy with the screws retained during removal.
- c) Connect the mains power to the display at the external isolator.
- d) Test the display from the CIS.
- e) Clean the display after all tests are complete and fill in the maintenance report, if required.

NOTE 1: For data termination on each line, insert jumper at links LK4 at the end of each line.

6.4.5 Removing a Logic Board

- a) Refer to **6.4.** and do steps 1 to 4.
- b) Make sure that there is no mains power at the LIVE terminal on the filter using a DMM.
- c) Locate the logic board and disconnect all plugs/sockets and screw terminals. Carefully note their locations for reconnection.
- d) Remove the nuts that attach the logic board to the equipment plate.
- e) Remove the board.

6.4.6 Installing a Logic Board

- a) Attach the new logic board with the nuts retained during removal.
- b) Check all switches and links on the new board are set to the same as those on the defective board that was removed.
- c) Connect the mains power to the display at the external isolator.
- d) Test the display from the CIS.
- e) Clean the display after all tests are complete and fill in the maintenance report, if required.

6.4.7 Removing the On Board PC

- g) Refer to **6.4.** and do steps 1 to 4.
- a) Locate the PC Card and disconnect the various connectors:
 - RS232 Cable
 - CAT 5 Ethernet cables
 - DC Power

Carefully note their locations for reconnection.

- b) Unscrew the fasteners that attach the board.
- c) Remove the board.

6.4.8 Installing the On Board PC

- a) Attach the new PC Card with the fasteners retained during removal.
- b) Connect the DC Power, RS232 and Ethernet cables.
- c) Connect the mains power to the display.
- d) Re-configure the PC Card.
- e) Test the display from the CIS.
- f) Clean the display after all tests are complete and fill in the maintenance report, if required.

6.4.9 Removing a LDR Board, PAS175DD

- a) Refer to **6.4.** and do steps 1 to 4.
- b) Make sure that there is no mains power at the LIVE terminal on the filter using a DMM.
- c) Locate the PAS175 board on the display plate assembly.
- d) Disconnect the molex type connector and note the correct position.
- e) Unscrew the 2 nuts that attach the PAS175 to the chassis and remove the Board.

6.4.10 Installing an LDR Board, PAS175DD

- a) Attach the new PAS175 to the display plate assembly using the nuts and washers retained during removal.
- b) Make sure that the panel does not block the face of the sensor from the light.
- c) Connect the molex type connector correctly.
- d) Connect the mains power to the display at the external isolator.
- e) Test the display from the CIS. During the test, place your hand over the PAS175 aperture and check the LED brightness decreases. Remove your hand and check the LED brightness increases.
- f) Clean the display after all tests are complete and fill in the maintenance report, if required.

6.4.11 Removing a Mains Filter

- a) Refer to **6.4.** and do steps 1 to 4.
- b) Make sure that there is no mains power at the LIVE terminal on the filter using a DMM.
- c) Disconnect the crimp terminals noting the locations. Leave the EARTH terminal until last.
- d) Unscrew the securing nuts that attach the filter and remove the filter. Retain all nuts and washers for further use.

6.4.12 Installing a Mains Filter

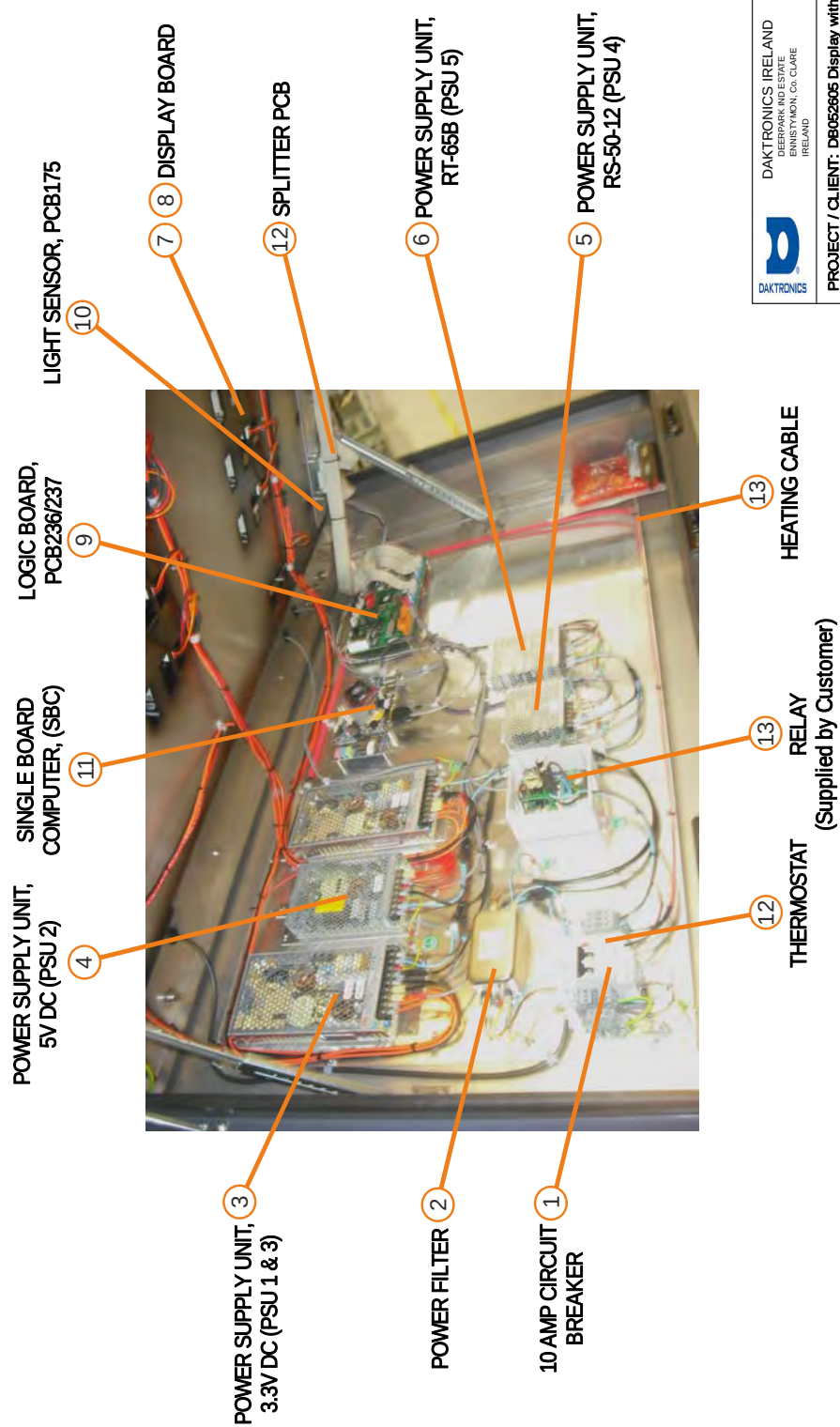
- a) Attach the new mains filter using the nuts and washers removed previously.
- b) Reconnect the EARTH connection and LIVE and NEUTRAL crimp terminals.
- c) Connect the mains power to the display at the external isolator.
- d) Use a DMM to check the LIVE output of the filter.
- e) Test the display from the CIS.
- f) Clean the display after all tests are complete and fill in the maintenance report, if required.

7 Display Wiring

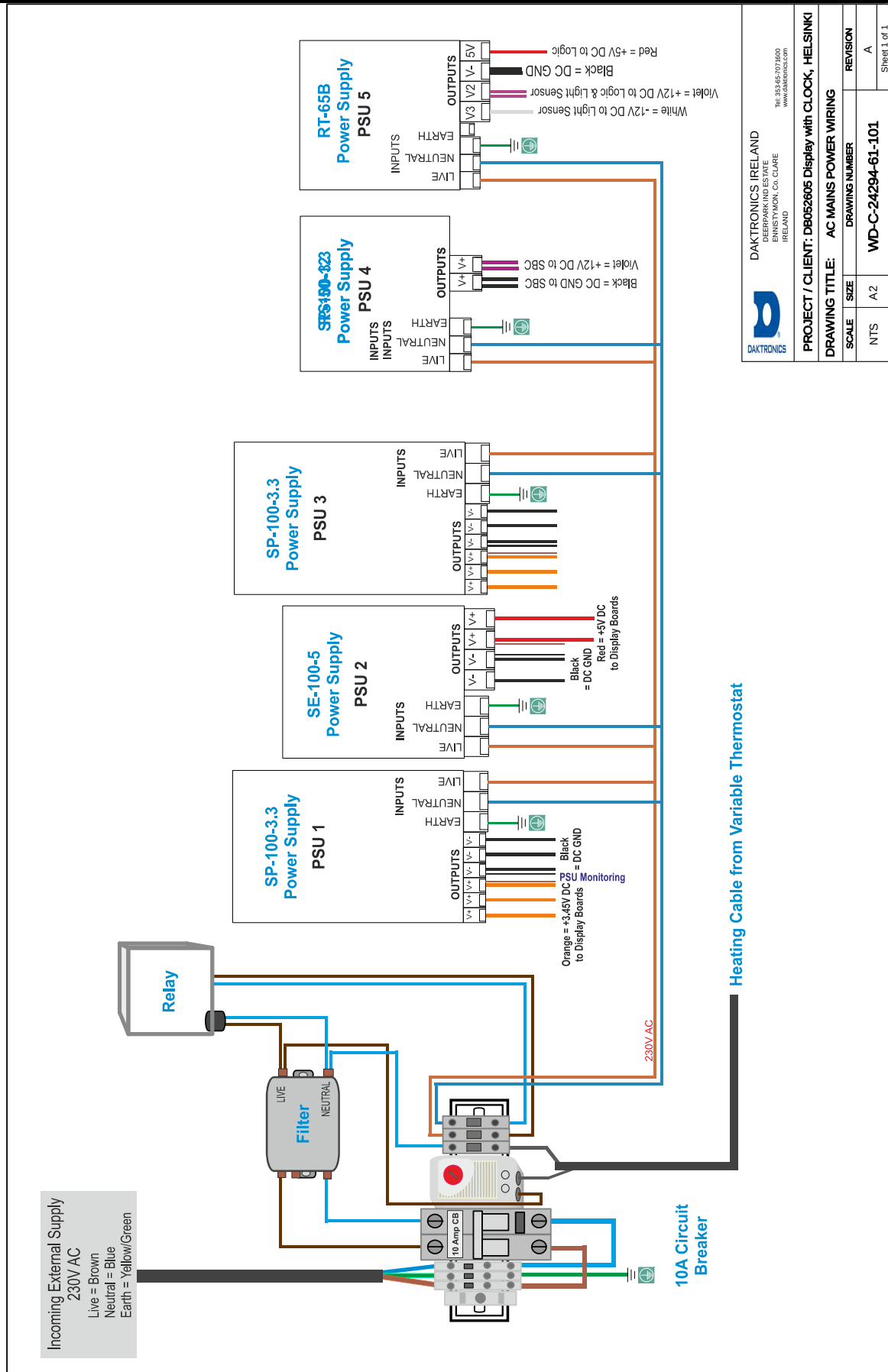
The following drawings show the internal wiring of the display.

Fig No	Title	Rev
WD-S-024294-61-100	Component Layout	A
WD-S-024294-61-101	Mains AC Wiring	A
WD-S-024294-61-102	COMMs Wiring	A
WD-S-024294-61-103	Data to Display Wiring	A

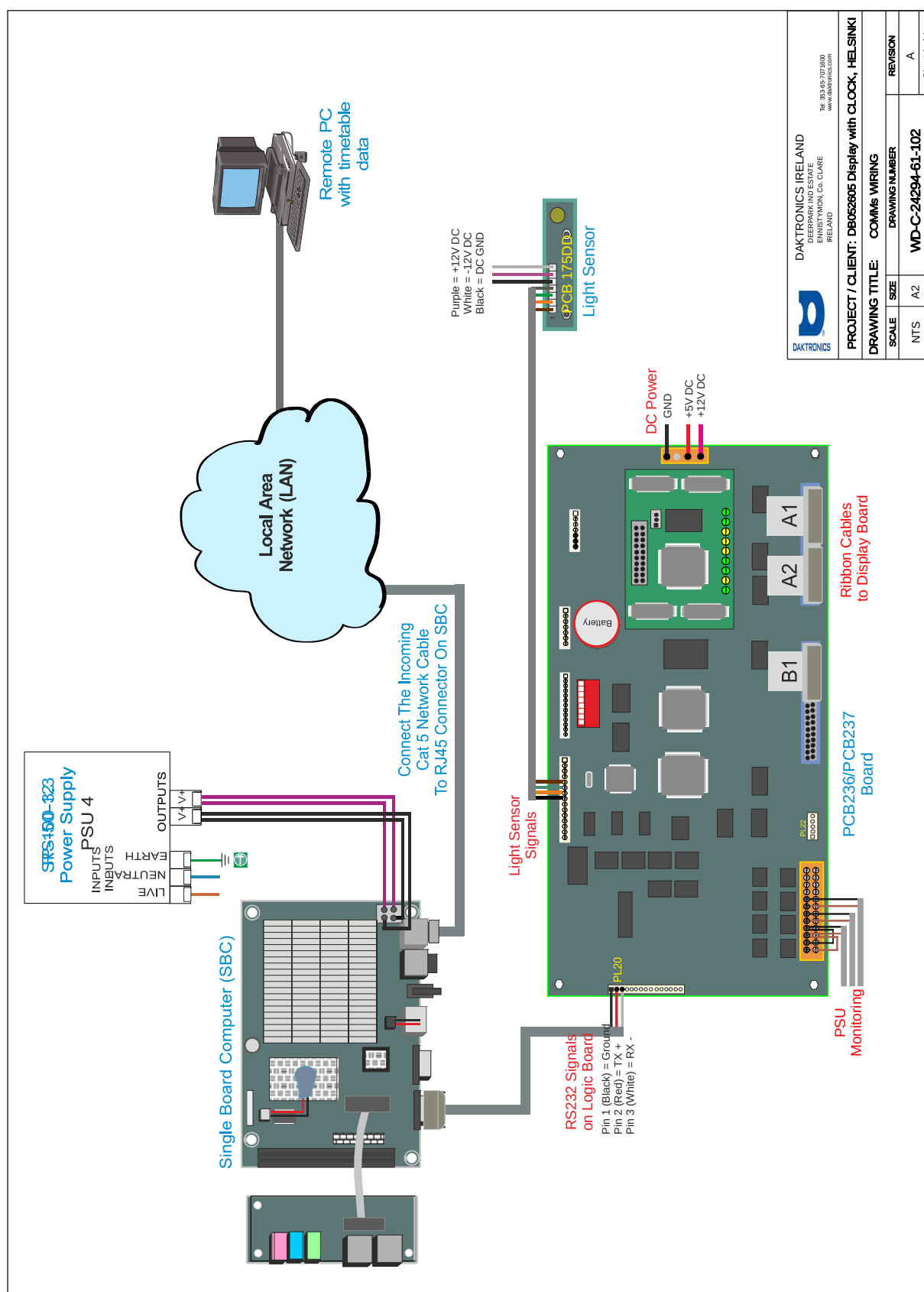
Table 7.1 – Wiring Diagrams.

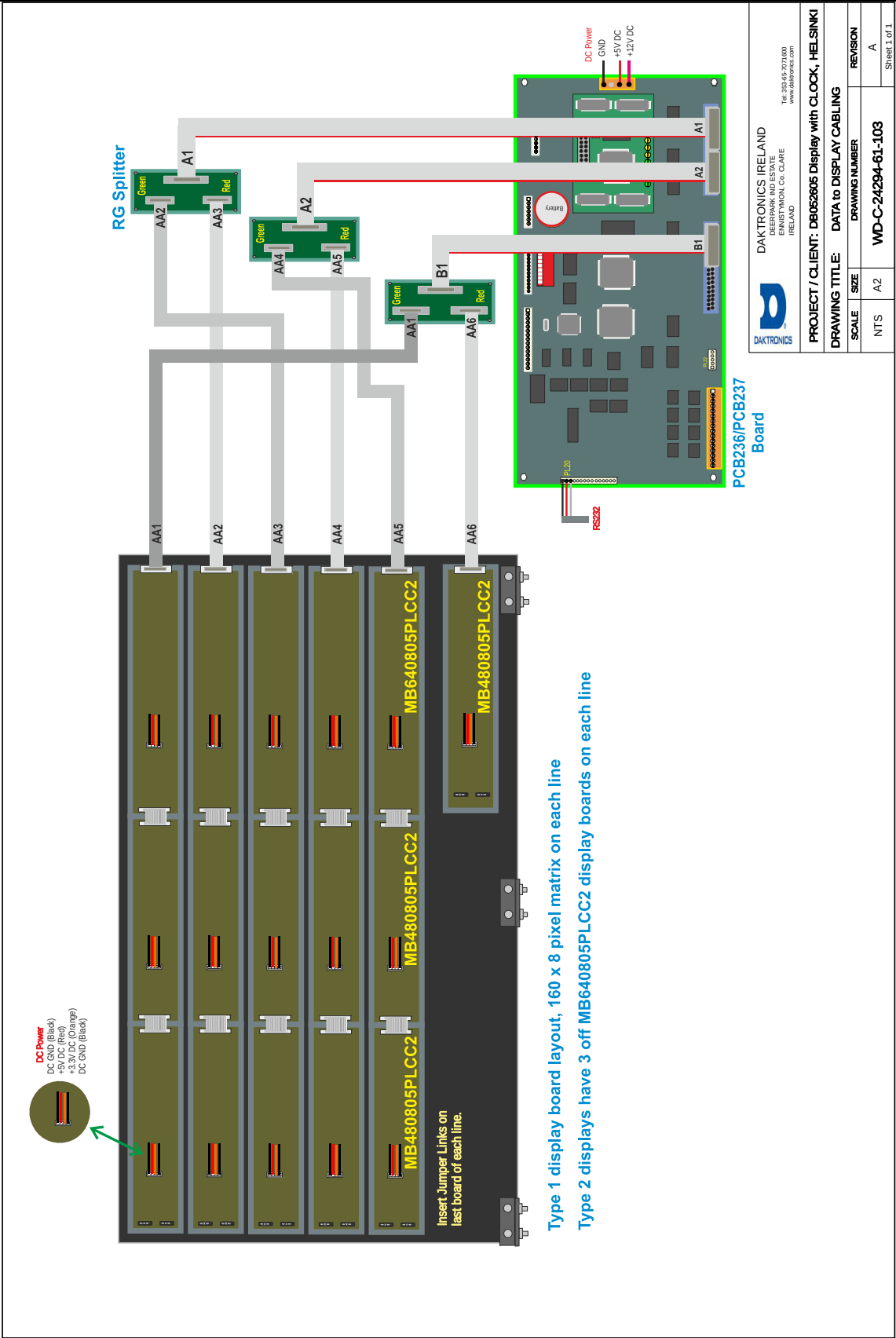


 DAKTRONICS		DAKTRONICS IRELAND DEERPARK INDUSTRIAL ESTATE ENNISTYMON, CO. CLARE IRELAND Tel: 353-65-7078600 www.daktronics.com	
PROJECT / CLIENT: DB062605 Display with CLOCK, HELSINKI			
DRAWING TITLE: COMPONENT LAYOUT			
SCALE	SIZE	DRAWING NUMBER	REVISION
NTS	A2	WD-C-24294-61-100	A
			Sheet 1 of 1



PROJECT / CLIENT: DB052605 Display with CLOAK, HELSINKI			
DRAWING TITLE: AC MAINS POWER WIRING			
SCALE	SIZE	DRAWING NUMBER	REVISION
NTS	A2	WD-C-24294-61-101	A
			Sheet 1 of 1





8 Appendix A – Write Filter Instructions

The ECM-A50M SBC is a 3.5" FDD size single board computer (SBC) that comes with a processor and various interfaces that allow the information to be extracted from the customer's data feed. The customer's data feed is connected to the RJ45 Network Socket LAN 2 port on the SBC.

Install TightVNCViewer on the maintenance PC/Laptop to connect to the Displays SBC. Logon to the SBC and do the following steps to enable or disable the write filter.

Enable EWF

1. To turn on the EWF, open Command Prompt, and type the following **"ewfmgr -all -enable"** (without the quotes and put a space either side of **-all**). Reboot again.
2. After the boot, open Command Prompt again, and type the following to check EWF status **"ewfmgr c:"**. It should say that the status is "ENABLED", among other information.

Disable EWF

1. To disable EWF so you can make changes to the OS. Reboot the SBC. At the command Prompt, type the following **"ewfmgr all: -commitanddisable -live"**.
2. To re-enable, type **"ewfmgr -all: -enable"**, which requires reboot.